

BG-002 — Gap-First Audit: Operator's "zero-defect / no-bluff / test-types / sub-submodule incorporation" Mandate vs Existing Governance

Date: 2026-09-19 · **Scope:** constitution/Constitution.md, constitution/CLAUDE.md, Lava root CLAUDE.md (§6.A–§6.AL), spot-checked submodule docs, constitution/submodules/*.

Executive summary

The operator's five-part demand is **already covered** — **mostly verbatim** — by the existing HelixConstitution anchor series and Lava's own Anti-Bluff Pact. Four of five demands are COVERED with near word-for-word matches to anchors already in force (§11.4.169, §11.4.262, §11.4.6, §11.4.261). The fifth (sub-submodule incorporation) is COVERED at the *rule* level (§11.4.28(C), §11.4.106, §11.4.207, §11.4.196(F)) but the underlying mechanisms are self-admittedly immature, and Lava's own operational scripts do not yet invoke any of constitution's seven nested engines — a real *compliance* gap, not a *governance-text* gap. Zero genuine textual gaps were found; no new anchor is proposed. Writing new rules on top of this corpus would itself be the bureaucratic-bloat/bluff the operator is trying to prevent.

(a) Mandatory creation of ALL supported test types for every piece of work

Verdict: COVERED. constitution/Constitution.md:9943 §11.4.169 ("Mandatory comprehensive test-type coverage with anti-bluff captured evidence") freezes a CLOSED 13-item test-type set — unit, integration, e2e, full-automation, Challenges, HelixQA, DDoS/load-flood,

security, stress+chaos, concurrency/atomicity, race-condition/deadlock, memory, benchmarking — driven to "as close to 100% coverage as the domain permits," each PASS backed by captured evidence, with only an honest §11.4.3 SKIP-with-reason (topology/hardware/credential genuinely absent) as the sole permitted absence. This is a near word-for-word restatement of the operator's own 2026-06-25 mandate quoted inline in the anchor. It is the strict enumerated expansion of §11.4.27 (No-Fakes-Beyond-Unit + 100%-Test-Type-Coverage). §11.4.224 additionally binds test-*first* (TDD) discipline + an 85% code-coverage floor to "ALL work... not only fixes," closing the adjacent gap the operator's "for every piece of work" phrase implies. Lava's root CLAUDE.md mirrors this at the project layer: the Anti-Bluff Pact's Seventh Law + §6.AE ("Comprehensive Challenge Coverage... all supported types of the tests") requires a per-feature Challenge for every module.

(b) Every test/tool run MUST produce machine-captured, deterministic evidence

Verdict: COVERED. The chain is §11.4.2 (Recorded-evidence requirement) → §11.4.5 (Captured-evidence quality analysis) → §11.4.69 (Universal Sink-Side Positive-Evidence Taxonomy) → §11.4.107 (self-validated AV/liveness oracles) → §11.4.115(F) (RED/GREEN are machine-written verdict files, never prose) → §11.4.226 (evidence-CLASS at closure, machine fields never a label) → **§11.4.262** ("Machine-created evidence at every gate"), which is the most directly on-point and reads almost as a restatement of the operator's exact words: *"every claim... MUST cite MACHINE-CREATED + MACHINE-VERIFIABLE evidence... Operator eyeballing / narrative-only PASS / 'looks fine' / 'no error' / prediction / guessing / assumption are FORBIDDEN."* It further requires every evidence-analyzer to be self-validated (golden-good/golden-bad/negative-control per §11.4.107(10)) so the *oracle itself* cannot bluff. Lava's Anti-Bluff Pact independently enforces the same thing project-side (§6.A real-binary contract tests, Seventh Law clause 1 Bluff-Audit stamps).

(c) Absolute prohibition on guessing, prediction, hedged language, fabricated results

Verdict: COVERED. constitution/Constitution.md:632 §11.4.6 ("No-guessing mandate") is a verbatim match: it forbids likely, probably, maybe, might, possibly, presumably, seems, appears to, guess, seemingly, apparently, perhaps, supposedly, conjectured and requires either a proven fact with captured evidence or an explicit UNCONFIRMED: / UNKNOWN: / PENDING_FORENSICS: marker with a tracked follow-up. This single anchor is cited as a load-bearing dependency by nearly every other anchor in the corpus (§11.4.102, §11.4.111, §11.4.150, §11.4.199, §11.4.201, §11.4.226, etc.), i.e. it is not an isolated rule but the corpus's own epistemic backbone. Lava's root CLAUDE.md restates it project-side under the same anchor number in the §6.AD.6 clause.

(d) Every constitution sub-submodule MUST be genuinely incorporated

Verdict: PARTIALLY-COVERED (rule-level COVERED; compliance-level gap, honestly self-tracked).

The *rule* exists: §11.4.28(C) is the explicit carve-out letting the constitution submodule host depth-1 reusable engines (constitution/submodules/<name>/, each with a helix-deps.yaml, zero nested own-org deps); §11.4.31 mandates the manifest + a Challenge proving a consumer can actually bootstrap and use the dependency; §11.4.106 mandates Docs Chain as *the* canonical doc/DB-sync engine ("consumed by reference... ad-hoc scripts are superseded and retired"); §11.4.207 mandates Continuum as *the* canonical resume engine. Crucially, §11.4.196(F) already generalizes the exact failure mode the operator fears — "CONFIGURED ≠ IN USE... claiming 'ready/in use' on config alone is a §11.4/§11.4.6 orchestration-layer bluff" — which is precisely "pinned-but-dormant" stated as a forbidden bluff class.

What is **not** true today (verified directly, not assumed): none of constitution's seven nested engines (anti_bluff, continuum, design-toolkit, docs_chain, helix_perf_cache, session_orchestrator, token_optimizer; clickup_sync exists only as design docs, not yet even a real git submodule) are invoked anywhere in Lava's own operational scripts (scripts/, docs/ sync tooling). They arrive transitively via

constitution/.gitmodules but are not in Lava's own .gitmodules, and grepping Lava's scripts/ tree found zero real invocations (only coincidental substring hits — e.g. anti-bluff-validate.sh is Lava's own script, unrelated to the anti_bluff submodule; docs mentioning "docs-chain" are design drafts about *future* adoption, not live usage). Several of the engines are self-described as immature (anti_bluff: "remaining mechanisms OWED"; design-toolkit: "FIRST INCREMENT"; helix_perf_cache: "Phase 0-1... skeleton"; session_orchestrator: "early scaffold... NOT implemented"; even docs_chain's own README states Phase 6 submodule-distribution is "PLANNED + OPERATOR-GATED," not shipped).

This is honestly self-tracked, not hidden: §11.4.227 ("Governance-corpus self-custody") is itself an anchor that measures the corpus's own gate-implementation rate and found 58% of named CM-* gates unimplemented — the corpus explicitly does not claim more machinery-maturity than it has. This is *correct* behavior per §11.4.6, not a gap. Note also: this is distinct from Lava's *own* 17-submodule fleet (Auth/Cache/Challenges/.../HelixQA), which Lava's own CLAUDE.md documents as fully wired with zero waivers — that fleet is a different, unrelated set of submodules from the constitution's internal sub-submodules the task asked about.

No new anchor proposed — the rule (genuine incorporation, no configured-but-unused bluffing) already exists via §11.4.28(C)/§11.4.106/§11.4.207/§11.4.196(F); what's missing is engineering time to wire the engines into Lava, which is an execution backlog item, not a governance-text gap.

(e) A standing, continuously-tracked zero-defect/zero-gap/zero-weak-spot bar

Verdict: COVERED, and correctly framed as the falsifiable/evidence-gated version, not the literal-impossible one.

constitution/Constitution.md:11390 §11.4.261 ("Zero-shortcomings/gaps/weak-spots/danger-zones invariant with a mechanical audit ratchet") is exactly on point and — importantly — already avoids the incoherent absolute the operator's phrasing risks. It mandates a mechanical audit sweep against a CLOSED, extensible vocabulary of finding classes, and requires the count to be **MONOTONE-DECREASING** from a declared brownfield baseline (the §11.4.66/§11.4.224(E) adoption-fence pattern) rather than an instantaneously-

true absolute; every finding must be either CLOSED (with evidence at the correct defect layer, per §11.4.226) or TRACKED (a real workable item with a regression guard, per §11.4.197/§11.4.135) — "silent absorption is a §11.4 PASS-bluff." It explicitly states "zero" is honest only *against the closed vocabulary* and that §11.4.118's discovery-pressure obligation remains in force (i.e., it does not claim omniscience). This composes with §11.4.239 (critical-invariant Definition of Done — failure-path scenarios as gates, not just happy-path), §11.4.250 (a heuristic tower signals a broken primitive rather than licensing another patch), and §11.4.266 (claim-vs-reality ledger — every *advertised* capability needs a passing challenge or it blocks release). Together these form exactly the "honest, falsifiable, continuously-audited" version of the zero-defect bar that real engineering can actually satisfy — a literal permanent zero is correctly *not* what's asserted anywhere in the corpus, and the operator's phrasing is satisfied by this ratchet-based framing rather than requiring a new, incoherent absolute claim.

Sub-submodule incorporation table (demand d detail)

Sub-submodule	What it does	Referenced by anchor	Used by Lava's own scripts?
anti_bluff	Mechanical anti-bluff seams (status-custody, RED/GREEN verdict wiring)	(mechanism underlying §11.4.146(D3)/§11.4.115(F))	No — self-described "remaining mechanisms OWED"
clickup_sync	ClickUp tracker sync (design-stage only)	none dedicated	No — not even a registered git submodule yet
continuum	Instant multi-stream resume engine (content-addressed)	§11.4.207	No — present transitively, not invoked

Sub-submodule	What it does	Referenced by anchor	Used by Lava's own scripts?
	snapshot store)		
design-toolkit	Reusable design-capability layer for OpenDesign	§11.4.162/ §11.4.220 (adjacent)	No — "FIRST INCREMENT... review-ready scaffold"
docs_chain	Canonical bidirectional doc/DB-sync engine	§11.4.106	No — Lava still uses its own ad-hoc sync scripts; docs_chain appears only in design drafts
helix_perf_cache	LLM inference-perf measurement/caching engine	none dedicated	No — "Phase 0-1 foundation... skeleton"
session_orchestrator	Alias-health registry + claim/scheduling engine	(feeds §11.4.187/ §11.4.196)	No — "early scaffold... NOT implemented"
token_optimizer	Token/cost-footprint minimizer for LLM pipelines	§11.4.141 (adjacent)	No — "active" upstream but unreferenced in Lava

Closing: genuine gaps found

Zero genuine textual gaps found. Every demand maps to an existing, often verbatim-matching anchor. The one area with a real shortfall — sub-submodule incorporation — is a *compliance/backlog* gap (Lava hasn't wired these engines in yet), not a *governance-text* gap: the rule already forbids exactly the "pinned but unused" failure mode (§11.4.196(F)), and the corpus already tracks its own implementation

debt honestly (§11.4.227). Proposing a new anchor here would either (a) duplicate §11.4.28(C)/§11.4.106/§11.4.207/§11.4.196(F), or (b) demand blanket forced-usage of every sub-engine regardless of applicability, which would itself violate §11.4.6's no-guessing/evidence-based-adoption spirit. No documentation edits were made to any governance file as part of this audit.